

Universidad Autónoma de Chihuahua

Facultad de Ingeniería



Titulo

Proyecto 2a. Evaluación

Materia

Base de Datos Avanzadas

Docente

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Fecha de entrega

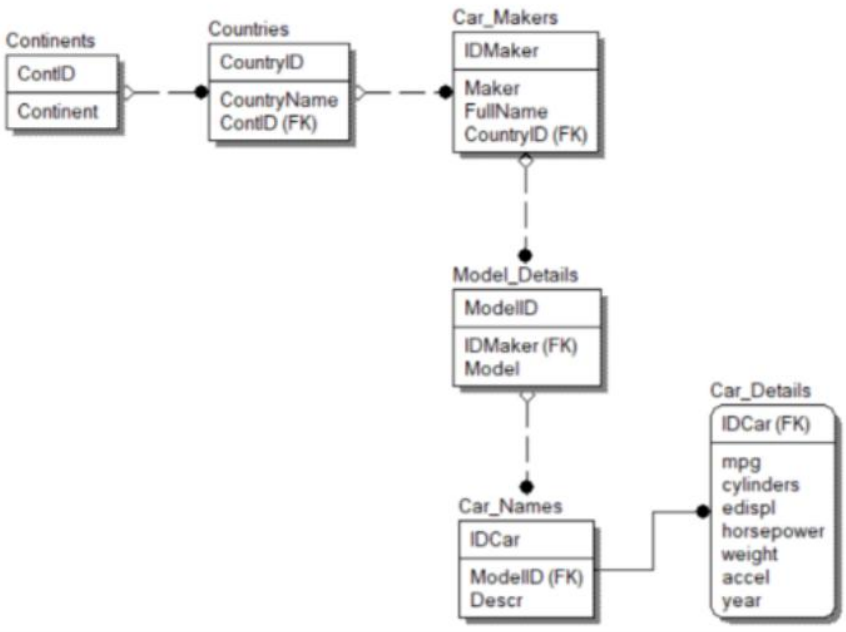
4 de noviembre de 2026

DOCUMENTACION DE APLICACIÓN EN PYTHON, UTILIZANDO MONGODB Y EL AGGRATE FRAMEWORK

Para desarrollar el proyecto, se nos dio un script de sql, para que pueda ser migrado a un script o json de MongoDB, recordando que, para usar MongoDB, no nos importa la estructura o constraints como en BD de Oracle, pero antes de empezar a agregar documentos a una Base de Datos de Mongo, debemos determinar cómo estructurar los documentos para poder realizar las consultas incluidas del reporte y justificar porque elegimos la siguiente estructura.

DISEÑO DE LA BASE DE DATOS DE MONGO

Primero tenemos que analizar el diseño de la base de datos relacional, ya se nos dio un diagrama de cómo se organiza:

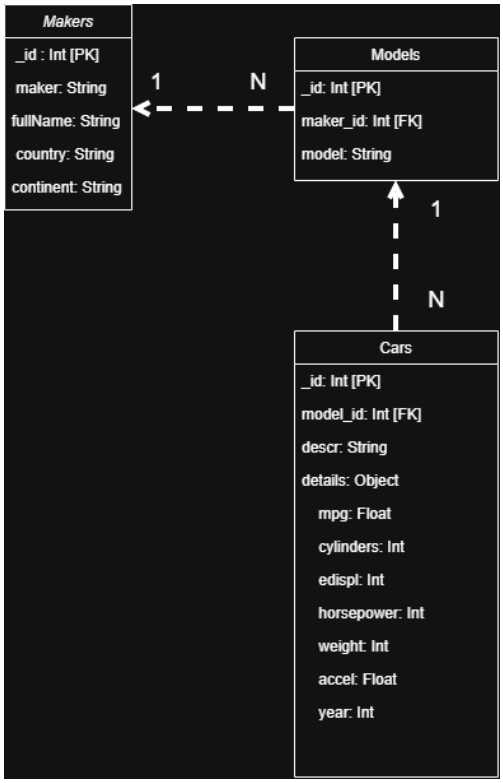


Para empezar a trabajar con MongoDB, nuestro objetivo es manejar documentos que sigan la estructura del script de SQL, pero recordemos que existen técnicas para el manejo de las relaciones en MongoDB, estos son los documentos embebidos y referenciados, con estas técnicas podremos trabajar el proyecto y sus consultas.

Para la migración se utilizó un diseño **híbrido (Embebido + Referenciado)** que reduce la necesidad de hacer uniones ($lookup) y agiliza las consultas:

- **Continents y Countries dentro de makers (Embebido):** Como Continents y Countries solo guardaban un nombre descriptivo, se incluyeron directamente como campos (country y continent) en cada fabricante. Esto evita hacer múltiples $lookup en reportes por país o continente.
- **Especificaciones dentro de cars (Embebido):** Dado que Car_Names y Car_Details tienen una relación estricta de 1 a 1, se unieron en la colección cars guardando la ficha técnica en el subdocumento details. Esto permite obtener la descripción y el rendimiento del auto en una sola lectura.
- **Marcas en models (Referenciado / Linking):** Se conservó la colección de marcas/líneas vinculada mediante maker_id para mantener el catálogo organizado y resolver consultas sobre fabricantes sin autos o con múltiples líneas sin duplicar datos.

Diagrama de cómo queda nuestra estructura de base de datos en MongoDB



De las 6 tablas que era en SQL, ahora solo manejamos 3 colecciones, ya que estamos implementando el manejo de documentos referenciados en models y el uso de documentos embebidos en cars y makers el cual se agregaron los campos directos de countries y continent, esto con la finalidad de que las consultas sean más fáciles de interpretar y evitar usar joins, que su equivalente en MongoDB es $lookup.

Colección makers

```
// ----------------------------------------
// 1. Colección: makers
// Embebe los datos geográficos (país y continente)
// ----------------------------------------
db.makers.insertMany([
  { _id: 1, maker: "gm", fullName: "General Motors", country: "usa", continent: "america" },
  { _id: 2, maker: "ford", fullName: "Ford Motor Company", country: "usa", continent: "america" },
  { _id: 3, maker: "stellantis", fullName: "Stellantis N.V.", country: "netherlands", continent: "europe" },
  { _id: 4, maker: "toyota", fullName: "Toyota Motor Corporation", country: "japan", continent: "asia" },
  { _id: 5, maker: "honda", fullName: "Honda Motor Co.", country: "japan", continent: "asia" },
  { _id: 6, maker: "nissan", fullName: "Nissan Motor Co.", country: "japan", continent: "asia" },
  { _id: 7, maker: "mazda", fullName: "Mazda Motor Corporation", country: "japan", continent: "asia" },
  { _id: 8, maker: "subaru", fullName: "Subaru Corporation", country: "japan", continent: "asia" },
  { _id: 9, maker: "mitsubishi", fullName: "Mitsubishi Motors", country: "japan", continent: "asia" },
  { _id: 10, maker: "suzuki", fullName: "Suzuki Motor Corporation", country: "japan", continent: "asia" },
  { _id: 11, maker: "hyundai", fullName: "Hyundai Motor Group", country: "korea", continent: "asia" },
  { _id: 12, maker: "volkswagen", fullName: "Volkswagen Group", country: "germany", continent: "europe" },
  { _id: 13, maker: "bmw", fullName: "BMW Group", country: "germany", continent: "europe" },
  { _id: 14, maker: "mercedes-benz", fullName: "Mercedes-Benz Group", country: "germany", continent: "europe" },
  { _id: 15, maker: "renault", fullName: "Renault Group", country: "france", continent: "europe" },
  { _id: 16, maker: "volvo", fullName: "Volvo Cars", country: "sweden", continent: "europe" },
  { _id: 17, maker: "jlr", fullName: "Jaguar Land Rover", country: "uk", continent: "europe" },
  { _id: 18, maker: "ferrari", fullName: "Ferrari N.V.", country: "italy", continent: "europe" },
  { _id: 19, maker: "saic", fullName: "SAIC Motor", country: "china", continent: "asia" },
  { _id: 20, maker: "chery", fullName: "Chery Automobile", country: "china", continent: "asia" }
]);
```

Colección models

```
// ----------------------------------------
// 2. Colección: models
// Referencia al fabricante vía maker_id (Linking)
// ----------------------------------------
db.models.insertMany([
  { _id: 1, maker_id: 1, model: "chevrolet" },
  { _id: 2, maker_id: 1, model: "gmc" },
  { _id: 3, maker_id: 1, model: "cadillac" },
  { _id: 4, maker_id: 1, model: "buick" },
  { _id: 5, maker_id: 2, model: "ford" },
  { _id: 6, maker_id: 2, model: "lincoln" },
  { _id: 7, maker_id: 3, model: "jeep" },
  { _id: 8, maker_id: 3, model: "dodge" },
  { _id: 9, maker_id: 3, model: "ram" },
  { _id: 10, maker_id: 3, model: "chrysler" },
  { _id: 11, maker_id: 3, model: "fiat" },
  { _id: 12, maker_id: 3, model: "peugeot" },
  { _id: 13, maker_id: 3, model: "alfa romeo" },
  { _id: 14, maker_id: 4, model: "toyota" },
  { _id: 15, maker_id: 4, model: "lexus" },
  { _id: 16, maker_id: 5, model: "honda" },
  { _id: 17, maker_id: 5, model: "acura" },
  { _id: 18, maker_id: 6, model: "nissan" },
  { _id: 19, maker_id: 6, model: "infiniti" },
  { _id: 20, maker_id: 7, model: "mazda" },
  { _id: 21, maker_id: 8, model: "subaru" },
  { _id: 22, maker_id: 9, model: "mitsubishi" },
  { _id: 23, maker_id: 10, model: "suzuki" },
  { _id: 24, maker_id: 11, model: "hyundai" },
  { _id: 25, maker_id: 11, model: "kia" },
  { _id: 26, maker_id: 11, model: "genesis" },
  { _id: 27, maker_id: 12, model: "volkswagen" },
  { _id: 28, maker_id: 12, model: "audi" },
  { _id: 29, maker_id: 12, model: "porsche" },
  { _id: 30, maker_id: 12, model: "lamborghini" },
  { _id: 31, maker_id: 12, model: "bentley" },
  { _id: 32, maker_id: 13, model: "bmw" },
  { _id: 33, maker_id: 13, model: "mini" },
  { _id: 34, maker_id: 14, model: "mercedes-benz" },
  { _id: 35, maker_id: 15, model: "renault" },
  { _id: 36, maker_id: 15, model: "dacia" },
  { _id: 37, maker_id: 16, model: "volvo" },
  { _id: 38, maker_id: 17, model: "land rover" },
  { _id: 39, maker_id: 17, model: "jaguar" },
  { _id: 40, maker_id: 18, model: "ferrari" },
  { _id: 41, maker_id: 19, model: "mg" },
  { _id: 42, maker_id: 20, model: "chery" }
]);
```

Colección cars

```
// ----------------------------------------
// 3. Colección: cars
// Unifica Car_Names con Car_Details (Embedding)
// Referencia a la marca/línea vía model_id (Linking)
// ----------------------------------------
db.cars.insertMany([
  { _id: 1, model_id: 1, descr: "chevrolet silverado 1500 lt 5.3", details: { mpg: 18, cylinders: 8, edispl: 325, horsepower: 355, weight: 5100, accel: 6.7, year: 2024 } },
  { _id: 2, model_id: 1, descr: "chevrolet tahoe lt 5.3", details: { mpg: 17, cylinders: 8, edispl: 325, horsepower: 355, weight: 5600, accel: 7, year: 2024 } },
  { _id: 3, model_id: 1, descr: "chevrolet equinox lt 1.5t", details: { mpg: 26, cylinders: 4, edispl: 91, horsepower: 175, weight: 3600, accel: 8.5, year: 2025 } },
  { _id: 4, model_id: 1, descr: "chevrolet trax 1rs 1.2t", details: { mpg: 30, cylinders: 3, edispl: 73, horsepower: 137, weight: 3100, accel: 9, year: 2025 } },
  { _id: 5, model_id: 1, descr: "chevrolet malibu lt 1.5t", details: { mpg: 32, cylinders: 4, edispl: 91, horsepower: 160, weight: 3150, accel: 8, year: 2024 } },
  { _id: 6, model_id: 1, descr: "chevrolet corvette stingray 6.2", details: { mpg: 19, cylinders: 8, edispl: 376, horsepower: 495, weight: 3650, accel: 2.9, year: 2025 } },
  { _id: 7, model_id: 1, descr: "chevrolet camaro ss 6.2", details: { mpg: 19, cylinders: 8, edispl: 376, horsepower: 455, weight: 3685, accel: 4, year: 2024 } },
  { _id: 8, model_id: 1, descr: "chevrolet colorado z71 2.7t", details: { mpg: 19, cylinders: 4, edispl: 165, horsepower: 310, weight: 4500, accel: 6.5, year: 2025 } },
  { _id: 9, model_id: 2, descr: "gmc sierra 1500 denali 6.2", details: { mpg: 17, cylinders: 8, edispl: 376, horsepower: 420, weight: 5600, accel: 5.9, year: 2024 } },
  { _id: 10, model_id: 2, descr: "gmc yukon denali 6.2", details: { mpg: 16, cylinders: 8, edispl: 376, horsepower: 420, weight: 5900, accel: 6.1, year: 2025 } },
  { _id: 11, model_id: 2, descr: "gmc terrain sle 1.5t", details: { mpg: 26, cylinders: 4, edispl: 91, horsepower: 175, weight: 3500, accel: 9.1, year: 2024 } },
  { _id: 12, model_id: 3, descr: "cadillac escalade 6.2", details: { mpg: 16, cylinders: 8, edispl: 376, horsepower: 420, weight: 5900, accel: 5.9, year: 2024 } },
  { _id: 13, model_id: 3, descr: "cadillac cts-v blackwing 6.2", details: { mpg: 15, cylinders: 8, edispl: 376, horsepower: 668, weight: 4125, accel: 3.6, year: 2024 } },
  { _id: 14, model_id: 3, descr: "cadillac xt5 premium luxury 2.0t", details: { mpg: 24, cylinders: 4, edispl: 122, horsepower: 235, weight: 4200, accel: 7.3, year: 2024 } },
  { _id: 15, model_id: 4, descr: "buick envista 1.2t", details: { mpg: 30, cylinders: 3, edispl: 73, horsepower: 137, weight: 3100, accel: 9.2, year: 2025 } },
  { _id: 16, model_id: 4, descr: "buick enclave 2.5t", details: { mpg: 22, cylinders: 4, edispl: 152, horsepower: 328, weight: 4650, accel: 6.5, year: 2025 } },
  { _id: 17, model_id: 5, descr: "ford f-150 xlt 2.7 ecoboost", details: { mpg: 21, cylinders: 6, edispl: 165, horsepower: 325, weight: 4700, accel: 6.3, year: 2024 } },
  { _id: 18, model_id: 5, descr: "ford f-150 raptor 3.5", details: { mpg: 16, cylinders: 6, edispl: 214, horsepower: 450, weight: 5700, accel: 5.4, year: 2024 } },
  { _id: 19, model_id: 5, descr: "ford mustang gt 5.0", details: { mpg: 18, cylinders: 8, edispl: 307, horsepower: 480, weight: 3900, accel: 4.2, year: 2024 } },
  { _id: 20, model_id: 5, descr: "ford mustang ecoboost 2.3", details: { mpg: 25, cylinders: 4, edispl: 140, horsepower: 315, weight: 3550, accel: 5.3, year: 2024 } },
  { _id: 21, model_id: 5, descr: "ford bronco sport big bend 1.5t", details: { mpg: 26, cylinders: 3, edispl: 91, horsepower: 180, weight: 3500, accel: 8.8, year: 2024 } },
  { _id: 22, model_id: 5, descr: "ford bronco badlands 2.7", details: { mpg: 17, cylinders: 6, edispl: 165, horsepower: 330, weight: 4750, accel: 6.4, year: 2024 } },
  { _id: 23, model_id: 5, descr: "ford ranger lariat 2.3", details: { mpg: 21, cylinders: 4, edispl: 140, horsepower: 270, weight: 4400, accel: 6.8, year: 2024 } },
  { _id: 24, model_id: 5, descr: "ford explorer st 3.0", details: { mpg: 20, cylinders: 6, edispl: 183, horsepower: 400, weight: 4700, accel: 5.2, year: 2025 } },
  { _id: 25, model_id: 5, descr: "ford escape st-line 1.5t", details: { mpg: 30, cylinders: 3, edispl: 91, horsepower: 180, weight: 3500, accel: 8.3, year: 2025 } },
  { _id: 26, model_id: 5, descr: "ford maverick xlt hybrid 2.5", details: { mpg: 37, cylinders: 4, edispl: 152, horsepower: 191, weight: 3700, accel: 7.7, year: 2024 } },
  { _id: 27, model_id: 6, descr: "lincoln navigator 3.5", details: { mpg: 18, cylinders: 6, edispl: 214, horsepower: 440, weight: 5900, accel: 6, year: 2024 } },
  { _id: 28, model_id: 6, descr: "lincoln aviator 3.0", details: { mpg: 21, cylinders: 6, edispl: 183, horsepower: 400, weight: 4900, accel: 5.6, year: 2024 } },
  { _id: 29, model_id: 7, descr: "jeep wrangler rubicon 3.6", details: { mpg: 19, cylinders: 6, edispl: 220, horsepower: 285, weight: 4500, accel: 7, year: 2024 } },
  { _id: 30, model_id: 7, descr: "jeep grand cherokee limited 3.6", details: { mpg: 22, cylinders: 6, edispl: 220, horsepower: 293, weight: 4600, accel: 7.5, year: 2024 } },
  { _id: 31, model_id: 7, descr: "jeep compass latitude 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 200, weight: 3600, accel: 7.5, year: 2024 } },
  { _id: 32, model_id: 7, descr: "jeep gladiator sport 3.6", details: { mpg: 19, cylinders: 6, edispl: 220, horsepower: 285, weight: 4900, accel: 7.5, year: 2024 } },
  { _id: 33, model_id: 8, descr: "dodge durango r/t 5.7", details: { mpg: 17, cylinders: 8, edispl: 348, horsepower: 360, weight: 5250, accel: 6.2, year: 2024 } },
  { _id: 34, model_id: 8, descr: "dodge charger scat pack 6.4", details: { mpg: 18, cylinders: 8, edispl: 392, horsepower: 485, weight: 4300, accel: 4.3, year: 2023 } },
  { _id: 35, model_id: 8, descr: "dodge challenger srt hellcat 6.2", details: { mpg: 16, cylinders: 8, edispl: 376, horsepower: 717, weight: 4450, accel: 3.6, year: 2023 } },
  { _id: 36, model_id: 8, descr: "dodge hornet gt 2.0t", details: { mpg: 24, cylinders: 4, edispl: 122, horsepower: 268, weight: 3700, accel: 6.5, year: 2024 } },
  { _id: 37, model_id: 9, descr: "ram 1500 laramie 3.0 hurricane", details: { mpg: 20, cylinders: 6, edispl: 183, horsepower: 420, weight: 5300, accel: 5.8, year: 2025 } },
  { _id: 38, model_id: 10, descr: "chrysler pacifica touring l 3.6", details: { mpg: 22, cylinders: 6, edispl: 220, horsepower: 287, weight: 4500, accel: 7.4, year: 2024 } },
  { _id: 39, model_id: 11, descr: "fiat 500x sport 1.3t", details: { mpg: 27, cylinders: 4, edispl: 79, horsepower: 177, weight: 3300, accel: 8.2, year: 2023 } },
  { _id: 40, model_id: 12, descr: "peugeot 2008 gt 1.2t", details: { mpg: 40, cylinders: 3, edispl: 73, horsepower: 130, weight: 2800, accel: 8.7, year: 2024 } },
  { _id: 41, model_id: 12, descr: "peugeot 3008 gt 1.6t", details: { mpg: 32, cylinders: 4, edispl: 98, horsepower: 165, weight: 3200, accel: 8.5, year: 2024 } },
  { _id: 42, model_id: 13, descr: "alfa romeo giulia 2.0t", details: { mpg: 27, cylinders: 4, edispl: 122, horsepower: 280, weight: 3500, accel: 5.5, year: 2024 } },
  { _id: 43, model_id: 13, descr: "alfa romeo stelvio quadrifoglio 2.9", details: { mpg: 19, cylinders: 6, edispl: 177, horsepower: 505, weight: 4400, accel: 3.6, year: 2024 } },
  { _id: 44, model_id: 14, descr: "toyota corolla le 2.0", details: { mpg: 35, cylinders: 4, edispl: 122, horsepower: 169, weight: 3100, accel: 8.3, year: 2024 } },
  { _id: 45, model_id: 14, descr: "toyota corolla hybrid le 1.8", details: { mpg: 50, cylinders: 4, edispl: 110, horsepower: 138, weight: 3150, accel: 9, year: 2024 } },
  { _id: 46, model_id: 14, descr: "toyota camry le hybrid 2.5", details: { mpg: 51, cylinders: 4, edispl: 152, horsepower: 225, weight: 3500, accel: 7.2, year: 2025 } },
  { _id: 47, model_id: 14, descr: "toyota rav4 xle 2.5", details: { mpg: 30, cylinders: 4, edispl: 152, horsepower: 203, weight: 3500, accel: 8, year: 2024 } },
  { _id: 48, model_id: 14, descr: "toyota rav4 hybrid xle 2.5", details: { mpg: 40, cylinders: 4, edispl: 152, horsepower: 219, weight: 3300, accel: 7.5, year: 2024 } },
  { _id: 49, model_id: 14, descr: "toyota tacoma trd off-road 2.4t", details: { mpg: 20, cylinders: 4, edispl: 146, horsepower: 278, weight: 4500, accel: 7, year: 2024 } },
  { _id: 50, model_id: 14, descr: "toyota tundra limited 3.4", details: { mpg: 20, cylinders: 6, edispl: 210, horsepower: 389, weight: 5600, accel: 6, year: 2024 } },
  { _id: 51, model_id: 14, descr: "toyota 4runner sr5 2.4t", details: { mpg: 20, cylinders: 4, edispl: 146, horsepower: 278, weight: 4700, accel: 7.5, year: 2025 } },
  { _id: 52, model_id: 14, descr: "toyota gr86 2.4", details: { mpg: 22, cylinders: 4, edispl: 146, horsepower: 228, weight: 2811, accel: 6.1, year: 2024 } },
  { _id: 53, model_id: 14, descr: "toyota gr supra 3.0", details: { mpg: 26, cylinders: 6, edispl: 183, horsepower: 382, weight: 3400, accel: 3.9, year: 2024 } },
  { _id: 54, model_id: 14, descr: "toyota hilux 2.7", details: { mpg: 24, cylinders: 4, edispl: 165, horsepower: 166, weight: 4100, accel: 9.5, year: 2024 } },
  { _id: 55, model_id: 15, descr: "lexus rx 350 2.4t", details: { mpg: 24, cylinders: 4, edispl: 146, horsepower: 275, weight: 4300, accel: 7.2, year: 2024 } },
  { _id: 56, model_id: 15, descr: "lexus es 300h 2.5", details: { mpg: 44, cylinders: 4, edispl: 152, horsepower: 215, weight: 3700, accel: 8, year: 2024 } },
  { _id: 57, model_id: 15, descr: "lexus lx 600 3.4", details: { mpg: 19, cylinders: 6, edispl: 210, horsepower: 409, weight: 5900, accel: 6.8, year: 2024 } },
  { _id: 58, model_id: 16, descr: "honda civic ex 1.5t", details: { mpg: 36, cylinders: 4, edispl: 91, horsepower: 180, weight: 3000, accel: 7.4, year: 2024 } },
  { _id: 59, model_id: 16, descr: "honda civic type r 2.0t", details: { mpg: 24, cylinders: 4, edispl: 122, horsepower: 315, weight: 3200, accel: 5, year: 2024 } },
  { _id: 60, model_id: 16, descr: "honda accord hybrid touring 2.0", details: { mpg: 44, cylinders: 4, edispl: 122, horsepower: 204, weight: 3500, accel: 6.9, year: 2024 } },
  { _id: 61, model_id: 16, descr: "honda cr-v ex 1.5t", details: { mpg: 30, cylinders: 4, edispl: 91, horsepower: 190, weight: 3500, accel: 8.4, year: 2024 } },
  { _id: 62, model_id: 16, descr: "honda hr-v lx 2.0", details: { mpg: 28, cylinders: 4, edispl: 122, horsepower: 158, weight: 3100, accel: 9.5, year: 2024 } },
  { _id: 63, model_id: 16, descr: "honda pilot trailsport 3.5", details: { mpg: 20, cylinders: 6, edispl: 214, horsepower: 285, weight: 4500, accel: 7, year: 2024 } },
  { _id: 64, model_id: 17, descr: "acura integra a-spec 1.5t", details: { mpg: 33, cylinders: 4, edispl: 91, horsepower: 200, weight: 3100, accel: 7, year: 2024 } },
  { _id: 65, model_id: 17, descr: "acura mdx 3.5", details: { mpg: 22, cylinders: 6, edispl: 214, horsepower: 290, weight: 4400, accel: 6.5, year: 2024 } },
  { _id: 66, model_id: 18, descr: "nissan versa sr 1.6", details: { mpg: 35, cylinders: 4, edispl: 98, horsepower: 122, weight: 2700, accel: 9.5, year: 2024 } },
```

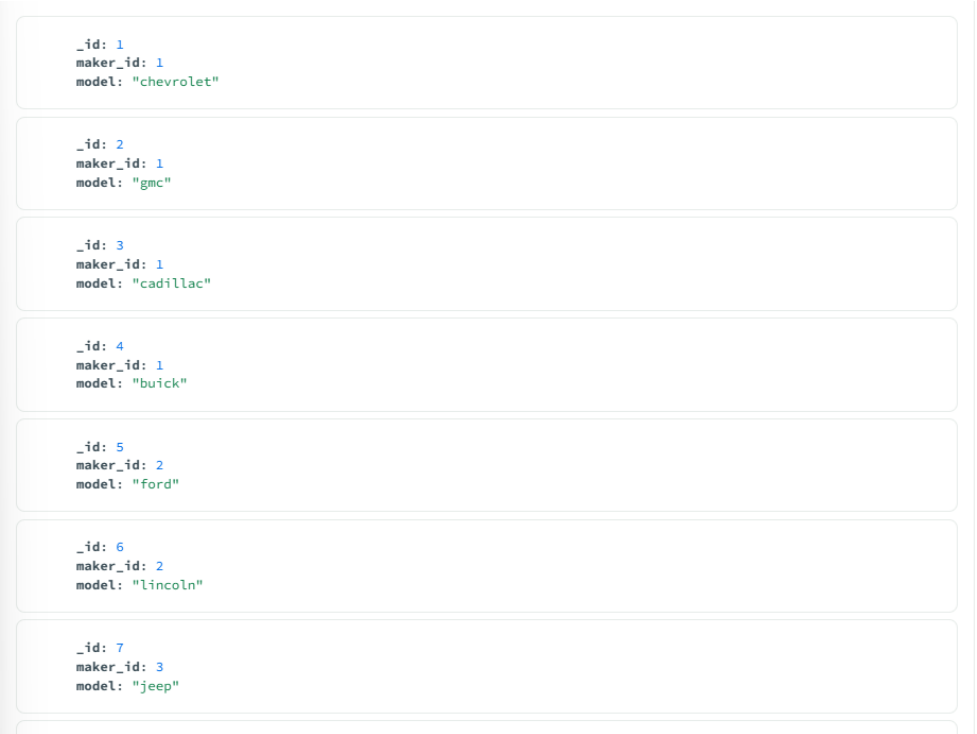
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  { _id: 67, model_id: 18, descr: "nissan sentra sv 2.0", details: { mpg: 34, cylinders: 4, edispl: 122, horsepower: 149, weight: 3050, accel: 8.7, year: 2024 } },
  { _id: 68, model_id: 18, descr: "nissan kicks sr 2.0", details: { mpg: 31, cylinders: 4, edispl: 122, horsepower: 141, weight: 3100, accel: 9.5, year: 2025 } },
  { _id: 69, model_id: 18, descr: "nissan rogue sv 1.5t", details: { mpg: 33, cylinders: 3, edispl: 91, horsepower: 201, weight: 3500, accel: 7.8, year: 2024 } },
  { _id: 70, model_id: 18, descr: "nissan frontier pro-4x 3.8", details: { mpg: 19, cylinders: 6, edispl: 232, horsepower: 310, weight: 4600, accel: 6.8, year: 2024 } },
  { _id: 71, model_id: 18, descr: "nissan z performance 3.0", details: { mpg: 22, cylinders: 6, edispl: 183, horsepower: 400, weight: 3500, accel: 4.3, year: 2024 } },
  { _id: 72, model_id: 18, descr: "nissan np300 2.5", details: { mpg: 23, cylinders: 4, edispl: 152, horsepower: 164, weight: 4000, accel: 11, year: 2024 } },
  { _id: 73, model_id: 19, descr: "infiniti q50 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 268, weight: 4000, accel: 7, year: 2025 } },
  { _id: 74, model_id: 20, descr: "mazda 3 hatchback 2.5 s", details: { mpg: 31, cylinders: 4, edispl: 152, horsepower: 191, weight: 3100, accel: 7.5, year: 2024 } },
  { _id: 75, model_id: 20, descr: "mazda cx-5 2.5 s", details: { mpg: 26, cylinders: 4, edispl: 152, horsepower: 187, weight: 3600, accel: 8.2, year: 2024 } },
  { _id: 76, model_id: 20, descr: "mazda cx-50 2.5 turbo", details: { mpg: 25, cylinders: 4, edispl: 152, horsepower: 256, weight: 3900, accel: 6.6, year: 2024 } },
  { _id: 77, model_id: 20, descr: "mazda cx-90 3.3 turbo", details: { mpg: 25, cylinders: 6, edispl: 201, horsepower: 280, weight: 4600, accel: 6.8, year: 2024 } },
  { _id: 78, model_id: 20, descr: "mazda mx-5 miata 2.0", details: { mpg: 30, cylinders: 4, edispl: 122, horsepower: 181, weight: 2350, accel: 5.7, year: 2024 } },
  { _id: 79, model_id: 21, descr: "subaru outback 2.5", details: { mpg: 29, cylinders: 4, edispl: 152, horsepower: 182, weight: 3700, accel: 8.8, year: 2024 } },
  { _id: 80, model_id: 21, descr: "subaru wrx 2.4t", details: { mpg: 22, cylinders: 4, edispl: 146, horsepower: 271, weight: 3400, accel: 5.5, year: 2024 } },
  { _id: 81, model_id: 21, descr: "subaru forester 2.5", details: { mpg: 29, cylinders: 4, edispl: 152, horsepower: 180, weight: 3500, accel: 8.5, year: 2025 } },
  { _id: 82, model_id: 21, descr: "subaru crosstrek 2.5", details: { mpg: 29, cylinders: 4, edispl: 152, horsepower: 182, weight: 3300, accel: 8.5, year: 2024 } },
  { _id: 83, model_id: 22, descr: "mitsubishi outlander se 2.5", details: { mpg: 26, cylinders: 4, edispl: 152, horsepower: 181, weight: 3600, accel: 8.6, year: 2024 } },
  { _id: 84, model_id: 22, descr: "mitsubishi mirage g4 1.2", details: { mpg: 39, cylinders: 3, edispl: 72, horsepower: 78, weight: 2100, accel: 11.5, year: 2024 } },
  { _id: 85, model_id: 22, descr: "mitsubishi l200 2.4", details: { mpg: 22, cylinders: 4, edispl: 146, horsepower: 181, weight: 4300, accel: 10.5, year: 2024 } },
  { _id: 86, model_id: 23, descr: "suzuki swift 1.2", details: { mpg: 44, cylinders: 3, edispl: 73, horsepower: 81, weight: 2100, accel: 12, year: 2024 } },
  { _id: 87, model_id: 23, descr: "suzuki vitara 1.4t", details: { mpg: 36, cylinders: 4, edispl: 85, horsepower: 138, weight: 2750, accel: 9.5, year: 2024 } },
  { _id: 88, model_id: 23, descr: "suzuki jimny 1.5", details: { mpg: 34, cylinders: 4, edispl: 91, horsepower: 101, weight: 2500, accel: 12, year: 2024 } },
  { _id: 89, model_id: 24, descr: "hyundai elantra sel 2.0", details: { mpg: 36, cylinders: 4, edispl: 122, horsepower: 147, weight: 2900, accel: 8.5, year: 2024 } },
  { _id: 90, model_id: 24, descr: "hyundai elantra n 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 276, weight: 3200, accel: 5.3, year: 2024 } },
  { _id: 91, model_id: 24, descr: "hyundai tucson sel 2.5", details: { mpg: 28, cylinders: 4, edispl: 152, horsepower: 187, weight: 3500, accel: 8.6, year: 2024 } },
  { _id: 92, model_id: 24, descr: "hyundai santa fe 2.5t", details: { mpg: 24, cylinders: 4, edispl: 152, horsepower: 277, weight: 4200, accel: 6.5, year: 2024 } },
  { _id: 93, model_id: 24, descr: "hyundai palisade 3.8", details: { mpg: 22, cylinders: 6, edispl: 232, horsepower: 291, weight: 4300, accel: 7.2, year: 2024 } },
  { _id: 94, model_id: 24, descr: "hyundai creta 1.5", details: { mpg: 35, cylinders: 4, edispl: 91, horsepower: 113, weight: 2700, accel: 11.5, year: 2024 } },
  { _id: 95, model_id: 25, descr: "kia forte lxs 2.0", details: { mpg: 33, cylinders: 4, edispl: 122, horsepower: 147, weight: 2900, accel: 8.6, year: 2024 } },
  { _id: 96, model_id: 25, descr: "kia sportage lx 2.5", details: { mpg: 28, cylinders: 4, edispl: 152, horsepower: 187, weight: 3500, accel: 8.6, year: 2024 } },
  { _id: 97, model_id: 25, descr: "kia telluride sx 3.8", details: { mpg: 23, cylinders: 6, edispl: 232, horsepower: 291, weight: 4400, accel: 7.1, year: 2024 } },
  { _id: 98, model_id: 25, descr: "kia k5 gt 2.5t", details: { mpg: 27, cylinders: 4, edispl: 152, horsepower: 290, weight: 3500, accel: 5.5, year: 2024 } },
  { _id: 99, model_id: 25, descr: "kia sorento 2.5", details: { mpg: 26, cylinders: 4, edispl: 152, horsepower: 191, weight: 3900, accel: 9, year: 2024 } },
  { _id: 100, model_id: 26, descr: "genesis g70 3.3t", details: { mpg: 20, cylinders: 6, edispl: 201, horsepower: 365, weight: 3900, accel: 4.5, year: 2024 } },
  { _id: 101, model_id: 26, descr: "genesis gv80 2.5t", details: { mpg: 22, cylinders: 4, edispl: 152, horsepower: 300, weight: 4700, accel: 6, year: 2024 } },
  { _id: 102, model_id: 27, descr: "volkswagen jetta se 1.5t", details: { mpg: 35, cylinders: 4, edispl: 91, horsepower: 158, weight: 3100, accel: 8, year: 2024 } },
  { _id: 103, model_id: 27, descr: "volkswagen golf gti 2.0t", details: { mpg: 28, cylinders: 4, edispl: 122, horsepower: 241, weight: 3200, accel: 5.9, year: 2024 } },
  { _id: 104, model_id: 27, descr: "volkswagen golf r 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 315, weight: 3500, accel: 4.5, year: 2024 } },
  { _id: 105, model_id: 27, descr: "volkswagen tiguan se 2.0t", details: { mpg: 26, cylinders: 4, edispl: 122, horsepower: 184, weight: 3800, accel: 8.5, year: 2024 } },
  { _id: 106, model_id: 27, descr: "volkswagen atlas 2.0t", details: { mpg: 22, cylinders: 4, edispl: 122, horsepower: 269, weight: 4500, accel: 7, year: 2024 } },
  { _id: 107, model_id: 27, descr: "volkswagen taos s 1.5t", details: { mpg: 31, cylinders: 4, edispl: 91, horsepower: 158, weight: 3200, accel: 7.9, year: 2024 } },
  { _id: 108, model_id: 28, descr: "audi a4 45 tfsi 2.0t", details: { mpg: 27, cylinders: 4, edispl: 122, horsepower: 261, weight: 3700, accel: 5.2, year: 2024 } },
  { _id: 109, model_id: 28, descr: "audi q5 45 tfsi 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 261, weight: 4100, accel: 5.7, year: 2024 } },
  { _id: 110, model_id: 28, descr: "audi rs 6 avant 4.0t", details: { mpg: 17, cylinders: 8, edispl: 244, horsepower: 591, weight: 4600, accel: 3.5, year: 2024 } },
  { _id: 111, model_id: 29, descr: "porsche 911 carrera 3.0", details: { mpg: 21, cylinders: 6, edispl: 183, horsepower: 379, weight: 3350, accel: 3.9, year: 2024 } },
  { _id: 112, model_id: 29, descr: "porsche cayenne 3.0", details: { mpg: 21, cylinders: 6, edispl: 183, horsepower: 348, weight: 4600, accel: 5.7, year: 2024 } },
  { _id: 113, model_id: 29, descr: "porsche macan t 2.0t", details: { mpg: 21, cylinders: 4, edispl: 122, horsepower: 261, weight: 4100, accel: 5.9, year: 2024 } },
  { _id: 114, model_id: 30, descr: "bmw 330i 2.0t", details: { mpg: 30, cylinders: 4, edispl: 122, horsepower: 255, weight: 3600, accel: 5.6, year: 2024 } },
  { _id: 115, model_id: 32, descr: "bmw m3 competition 3.0", details: { mpg: 19, cylinders: 6, edispl: 183, horsepower: 503, weight: 3900, accel: 3.8, year: 2024 } },
  { _id: 116, model_id: 32, descr: "bmw x3 xdrive30i 2.0t", details: { mpg: 26, cylinders: 4, edispl: 122, horsepower: 255, weight: 4200, accel: 6, year: 2025 } },
  { _id: 117, model_id: 32, descr: "bmw x5 xdrive40i 3.0t", details: { mpg: 22, cylinders: 6, edispl: 183, horsepower: 375, weight: 4900, accel: 5.5, year: 2024 } },
  { _id: 118, model_id: 33, descr: "mini cooper s 2.0t", details: { mpg: 31, cylinders: 4, edispl: 122, horsepower: 201, weight: 2900, accel: 6.4, year: 2025 } },
  { _id: 119, model_id: 34, descr: "mercedes-benz c 300 2.0t", details: { mpg: 29, cylinders: 4, edispl: 122, horsepower: 255, weight: 3800, accel: 5.9, year: 2024 } },
  { _id: 120, model_id: 34, descr: "mercedes-benz e 350 2.0t", details: { mpg: 27, cylinders: 4, edispl: 122, horsepower: 255, weight: 4100, accel: 6.2, year: 2024 } },
  { _id: 121, model_id: 34, descr: "mercedes-benz gle 350 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 255, weight: 4300, accel: 6.6, year: 2024 } },
  { _id: 122, model_id: 34, descr: "mercedes-benz amg g 63 4.0", details: { mpg: 14, cylinders: 8, edispl: 244, horsepower: 577, weight: 5800, accel: 4.4, year: 2024 } },
  { _id: 123, model_id: 35, descr: "renault duster 1.6", details: { mpg: 32, cylinders: 4, edispl: 98, horsepower: 113, weight: 2600, accel: 12, year: 2024 } },
  { _id: 124, model_id: 35, descr: "renault kardian 1.0t", details: { mpg: 40, cylinders: 3, edispl: 61, horsepower: 99, weight: 2600, accel: 11, year: 2025 } },
  { _id: 125, model_id: 37, descr: "volvo xc60 b5 2.0t", details: { mpg: 25, cylinders: 4, edispl: 122, horsepower: 247, weight: 4200, accel: 6.4, year: 2024 } },
  { _id: 126, model_id: 37, descr: "volvo xc90 b6 2.0t", details: { mpg: 22, cylinders: 4, edispl: 122, horsepower: 295, weight: 4700, accel: 6.1, year: 2024 } },
  { _id: 127, model_id: 37, descr: "volvo s60 b5 2.0t", details: { mpg: 28, cylinders: 4, edispl: 122, horsepower: 247, weight: 3700, accel: 6.3, year: 2024 } },
  { _id: 128, model_id: 38, descr: "land rover defender 3.0", details: { mpg: 19, cylinders: 6, edispl: 183, horsepower: 395, weight: 5100, accel: 5.7, year: 2024 } },
  { _id: 129, model_id: 38, descr: "land rover range rover sport 3.0", details: { mpg: 20, cylinders: 6, edispl: 183, horsepower: 355, weight: 5000, accel: 5.7, year: 2024 } },
  { _id: 130, model_id: 39, descr: "jaguar f-pace p250 2.0t", details: { mpg: 24, cylinders: 4, edispl: 122, horsepower: 246, weight: 4100, accel: 6.9, year: 2024 } },
  { _id: 131, model_id: 39, descr: "jaguar f-type r 5.0", details: { mpg: 18, cylinders: 8, edispl: 305, horsepower: 575, weight: 3900, accel: 3.5, year: 2024 } },
  { _id: 132, model_id: 40, descr: "ferrari roma 3.9", details: { mpg: 17, cylinders: 8, edispl: 238, horsepower: 612, weight: 3460, accel: 3.4, year: 2024 } },
  { _id: 133, model_id: 40, descr: "ferrari 812 competizione 6.5", details: { mpg: 14, cylinders: 12, edispl: 397, horsepower: 819, weight: 3600, accel: 2.8, year: 2023 } },
  { _id: 134, model_id: 41, descr: "mg zs 1.5", details: { mpg: 34, cylinders: 4, edispl: 91, horsepower: 106, weight: 2900, accel: 12, year: 2024 } },
  { _id: 135, model_id: 41, descr: "mg 5 1.5", details: { mpg: 38, cylinders: 4, edispl: 91, horsepower: 111, weight: 2700, accel: 11.5, year: 2024 } },
  { _id: 136, model_id: 42, descr: "mg hs 1.5t", details: { mpg: 30, cylinders: 4, edispl: 91, horsepower: 168, weight: 3400, accel: 9.5, year: 2024 } },
  { _id: 137, model_id: 42, descr: "chery tiggo 4 pro 1.5", details: { mpg: 30, cylinders: 4, edispl: 91, horsepower: 113, weight: 3000, accel: 12, year: 2024 } },
  { _id: 138, model_id: 42, descr: "chery tiggo 7 pro 1.5t", details: { mpg: 28, cylinders: 4, edispl: 91, horsepower: 154, weight: 3300, accel: 10, year: 2024 } }
]);
```

En total, se tiene la siguiente cantidad de documentos para cada colección:

Makers: tiene 20 documentos



Models: tiene 42 documentos



Cars: tiene 138 documentos

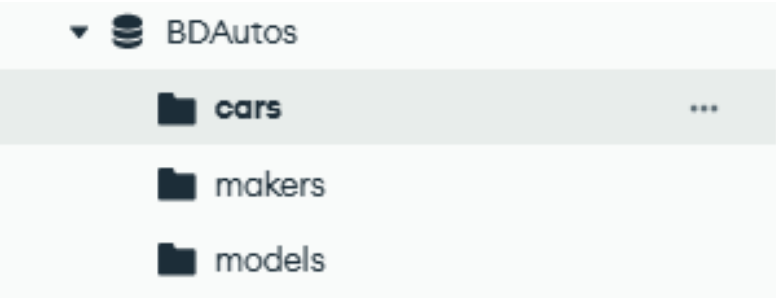

```
_id: 1
model_id: 1
descr: "chevrolet silverado 1500 lt 5.3"
▾ details: Object (7)
    mpg: 18
    cylinders: 8
    edispl: 325
    horsepower: 355
    weight: 5100
    accel: 6.7
    year: 2024

_id: 2
model_id: 1
descr: "chevrolet tahoe lt 5.3"
▾ details: Object (7)
    mpg: 17
    cylinders: 8
    edispl: 325
    horsepower: 355
    weight: 5600
    accel: 7
    year: 2024

_id: 3
model_id: 1
descr: "chevrolet equinox lt 1.5t"
▾ details: Object (7)
    mpg: 26
    cylinders: 4
    edispl: 91
    horsepower: 175
    weight: 3600
    accel: 8.5
    year: 2025
```


Aquí observamos los documentos embebidos y los referenciados.

El anterior fue el código para insertar todo los documentos a la base de datos BDAutos, como se tiene guardado estas 3 colecciones, ya que hemos migrado los datos a MongoDB, podemos explicar la aplicación en Python.



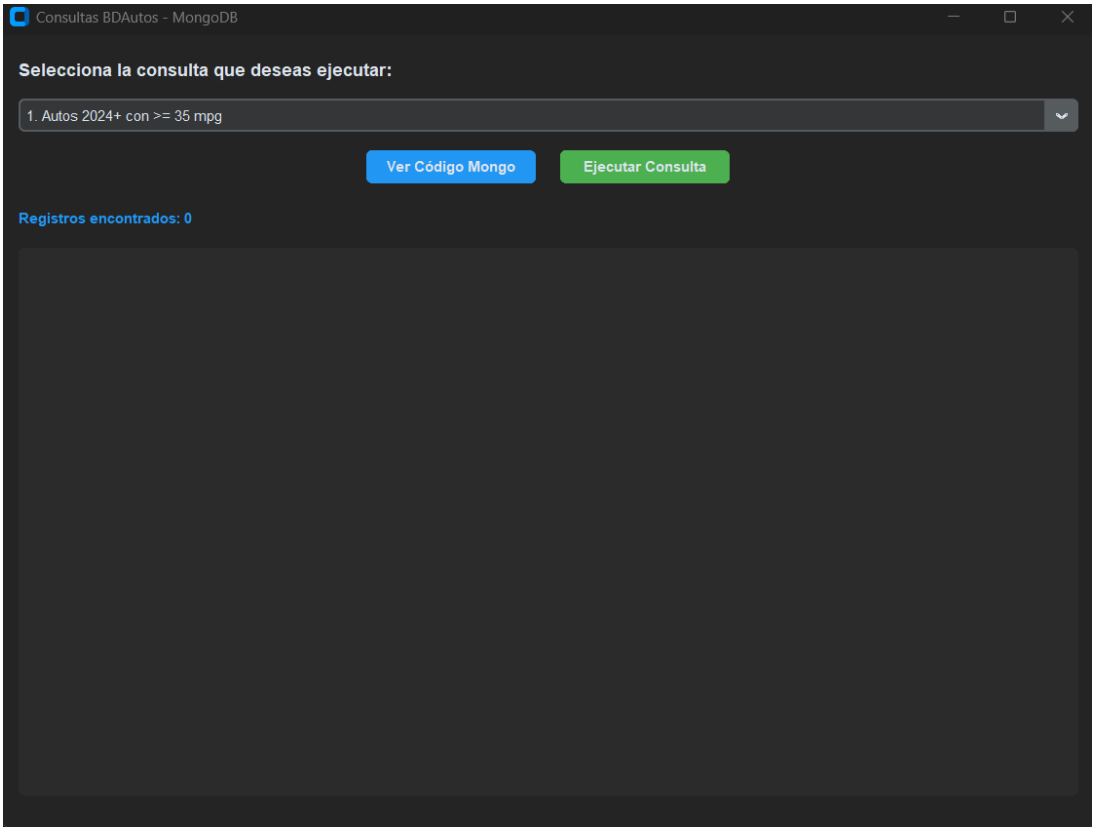
EJECUCIONES DE LA APLICACIÓN EN PYTHON

Para poder utilizar MongoDB requerimos de importar una librería **pymongo**, y realizamos la conexión a la base de datos “BDAutos” para poder realizar la consultas correspondientes.

```
import pymongo
```

```
# ========================================
# 1. CONEXIÓN A LA BASE DE DATOS
# ========================================
try:
    cliente = pymongo.MongoClient("mongodb://localhost:27017/", serverSelectionTimeoutMS=2000)
    db = cliente["BDAutos"]
    cliente.server_info()
except Exception as e:
    print(f"Error al conectar a MongoDB: {e}")
```

La correr el programa, tenemos la siguiente interfaz que fue creada utilizando la librería de customtkinter.



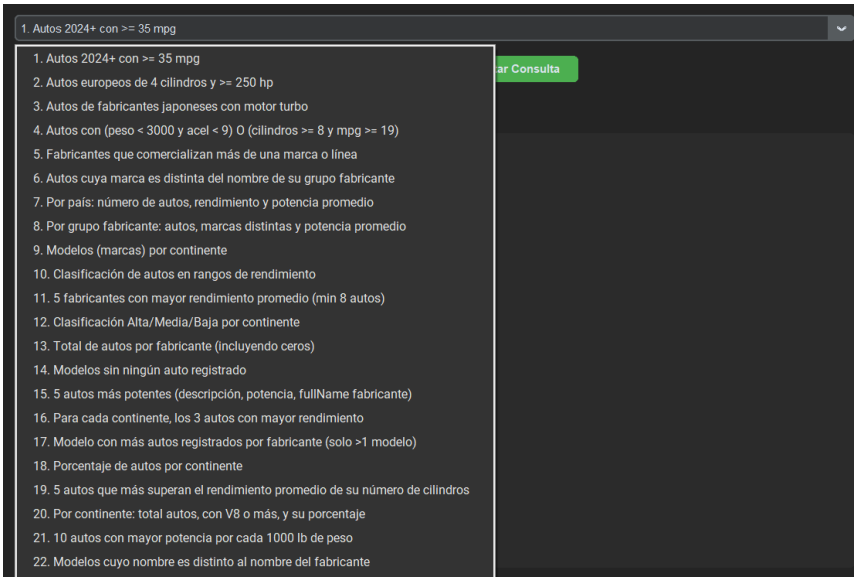
Para cumplir con los requisitos del proyecto, se creo un combobox que contiene las 25 consultas que se encuentra en el repositorio en su propio archivo, para poder utilizar las consultas en Python, requerimos usar del framework aggregate.

```
CONSULTAS = {
    "1. Autos 2024+ con >= 35 mpg": (
        "cars", [
            { "$match": { "details.year": { "$gte": 2024 }, "details.mpg": { "$gte": 35 } } },
            { "$project": { "_id": 0, "descripcion": "$descr", "año": "$details.year", "rendimiento": "$details.mpg" } },
            { "$sort": { "rendimiento": -1 } }
        ]
    ),
    "2. Autos europeos de 4 cilindros y >= 250 hp": (
        "cars", [
            { "$match": { "details.cylinders": 4, "details.horsepower": { "$gte": 250 } } },
            { "$lookup": { "from": "models", "localField": "model_id", "foreignField": "_id", "as": "model_info" } },
            { "$unwind": "$model_info" },
            { "$lookup": { "from": "makers", "localField": "model_info.maker_id", "foreignField": "_id", "as": "maker_info" } },
            { "$unwind": "$maker_info" },
            { "$match": { "maker_info.continent": "europe" } },
            { "$project": { "_id": 0, "descripcion": "$descr", "pais": "$maker_info.country", "potencia": "$details.horsepower" } },
            { "$sort": { "potencia": -1 } }
        ]
    ),
```

Guardamos las consultas en un diccionario, su llave es el string de la consulta en el combobox y su valor es la consulta que contiene la colección donde empieza la instrucción y un arreglo que contiene el pipeline, mejor dicho, las **instrucciones del pipeline** para poder utilizar la función de aggregate que viene con la librería de pymongo.

```
coleccion, pipeline = CONSULTAS[seleccion]
resultado_cursor = db[coleccion].aggregate(pipeline)
resultados = list(resultado_cursor)
```

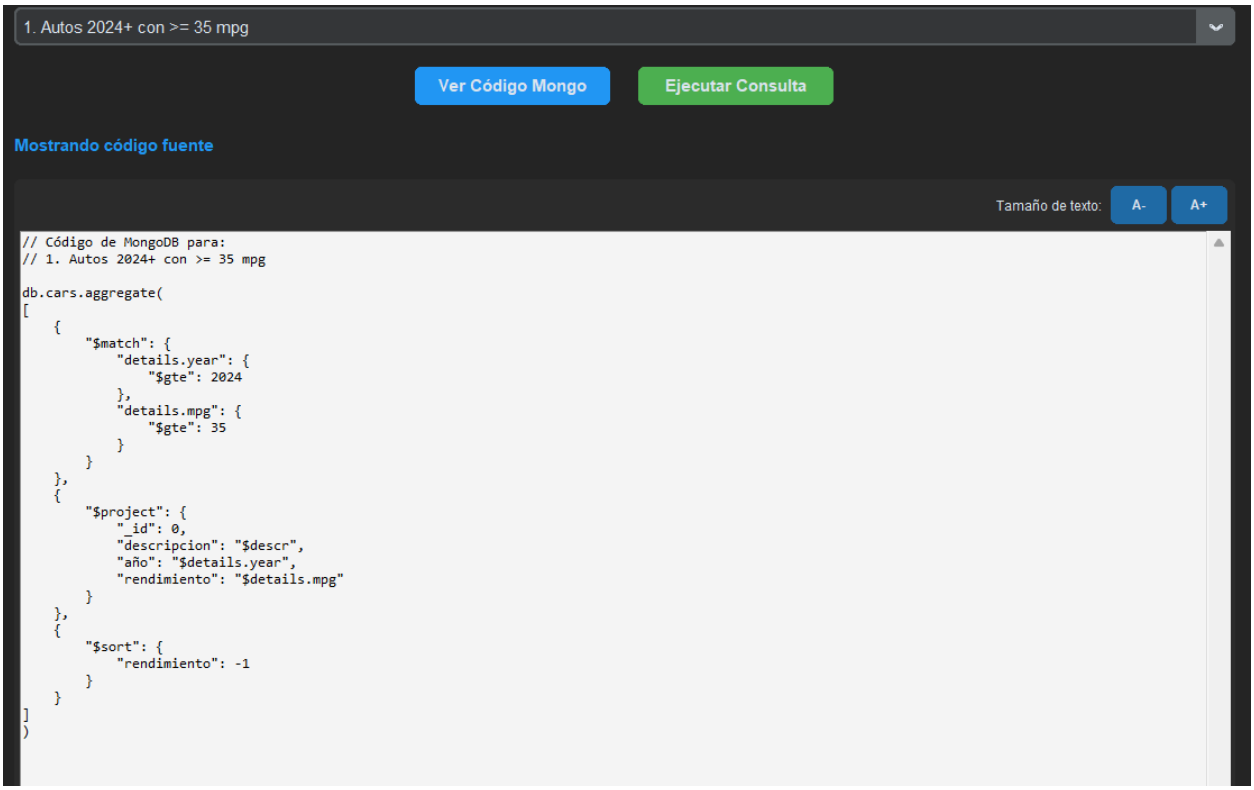
Para usar el aggregate tenemos que obtener la consulta, selección es la opción que tomemos del combobox



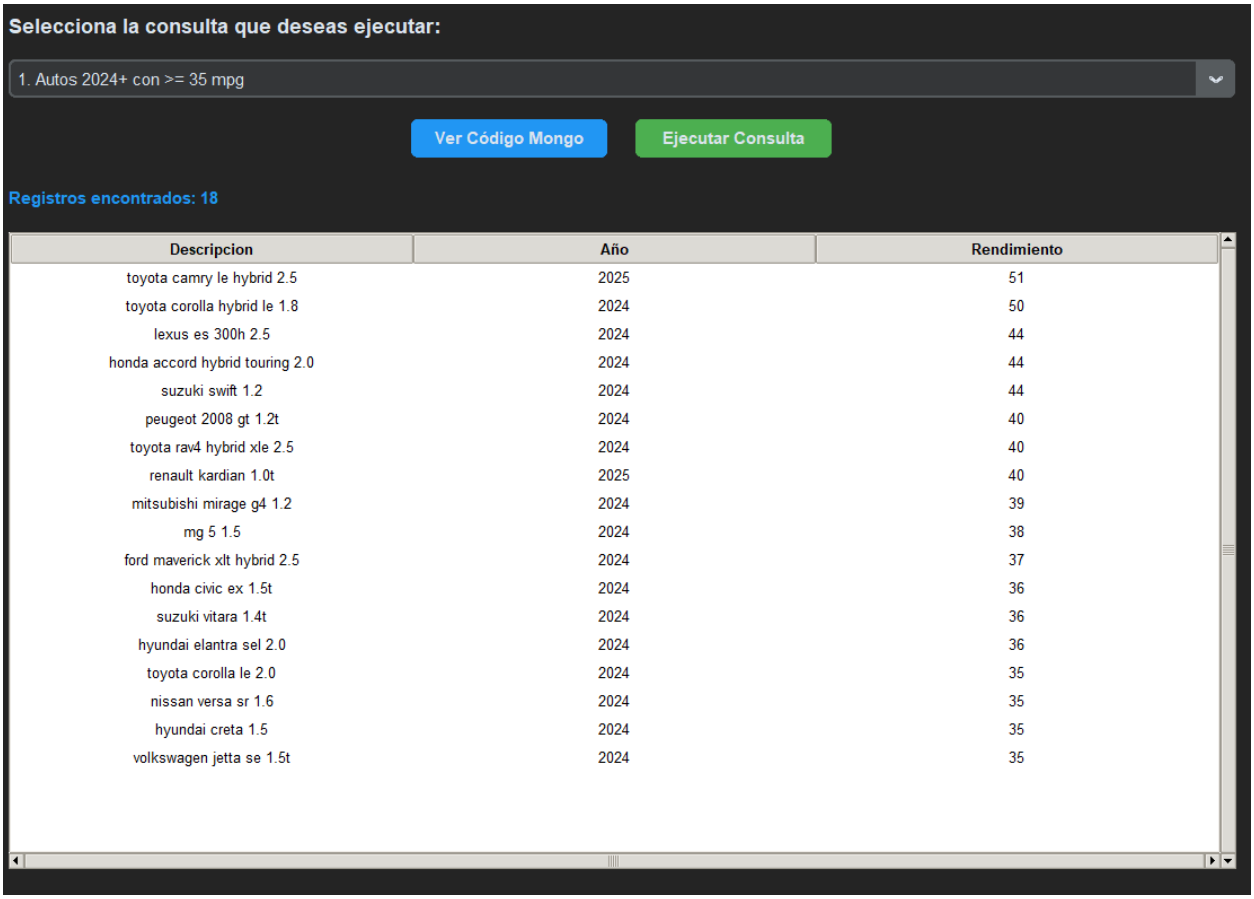
Después esa consulta obtenemos el pipeline y su correspondiente colección, para luego mandar a llamar la función.

```
resultado_cursor = db[coleccion].aggregate(pipeline)
```

Esta es la manera que el programa de comunica con MongoDB y obtenemos respuesta.



La primera parte del proyecto es que muestre el código de la consulta, en este caso como seria verlo desde mongosh o compass para luego ser ejecutado con la función anterior, de esta manera sabemos que nuestra consulta fue recibida por MongoDB y retorna los valores solicitados.



La ejecución se presenta de dos maneras, primero la respuesta si se puede interpretar de forma tabular, en pocas palabras que la consulta no contenga datos anidados o documentos embebidos, tendremos 2 formas de visualizar las ejecuciones, una que es la forma de tabla y la segunda es la forma que lo hubiera regresado compass, en forma de documento.

Por ejemplo, la consulta 22, nos enseña la segunda forma de visualizar la ejecución:

Selecciona la consulta que deseas ejecutar:

22. Modelos cuyo nombre es distinto al nombre del fabricante

Ver Código Mongo | Ejecutar Consulta

Mostrando código fuente

Tamaño de texto: A- A+

```
db.makers.aggregate(
[
    {
        "$lookup": {
            "from": "models",
            "localField": "_id",
            "foreignField": "maker_id",
            "as": "models_list"
        }
    },
    {
        "$project": {
            "fabricante": "$maker",
            "modelos_filtrados": {
                "$filter": {
                    "input": "$models_list",
                    "as": "m",
                    "cond": {
                        "$ne": [
                            "$$m.model",
                            "$maker"
                        ]
                    }
                }
            }
        }
    },
    {
        "$match": {
            "modelos_filtrados.0": {
                "$exists": true
            }
        }
    },
    {
        "$project": {
            "_id": 0,
            "fabricante": 1,
            "modelos_distintos": "$modelos_filtrados.model"
        }
    }
]
```

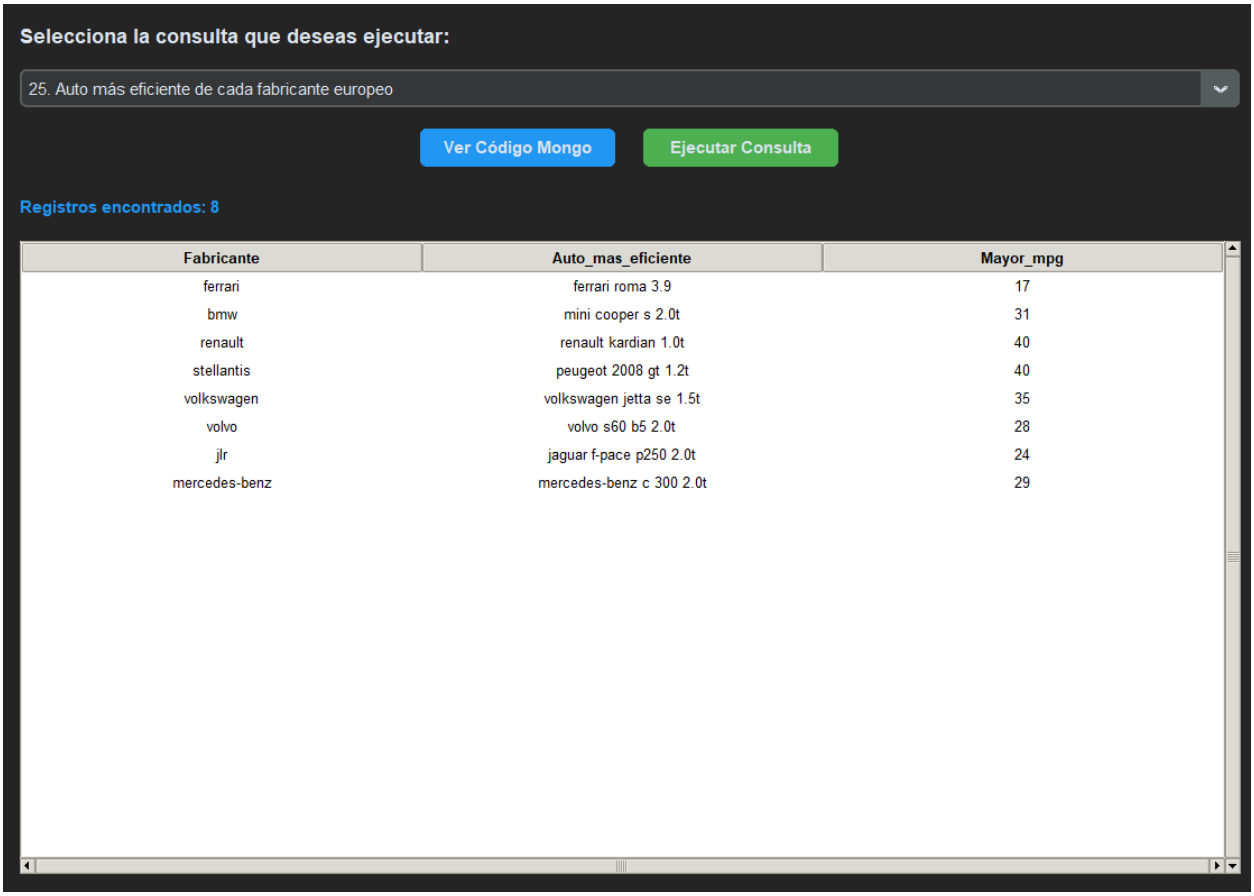



```
},
{
    "fabricante": "honda",
    "modelos_distintos": [
        "acura"
    ]
},
{
    "fabricante": "nissan",
    "modelos_distintos": [
        "infiniti"
    ]
},
{
    "fabricante": "hyundai",
    "modelos_distintos": [
        "kia",
        "genesis"
    ]
},
{
    "fabricante": "volkswagen",
    "modelos_distintos": [
        "audi",
        "porsche",
        "lamborghini",
        "bentley"
    ]
},
{
    "fabricante": "bmw",
    "modelos_distintos": [
        "mini"
```


Vemos que tenemos datos que contiene más de un dato o un objeto, por eso no podemos mostrarlo de forma tabular.

Conclusión

Esta aplicación muestra al usuario las opciones de ver el código de la consulta y el resultado de ejecutarla directamente en mongosh o compass, es más fácil visualizar y sobre todo porque gracias al framework Aggregate nos permite realizar consulta mucho más especificas ya que al trabajar con un modelo NoSQL estamos muy limitados a usar operaciones como joins o lookups, o hacer consultas simultáneas con facet.



ENLACE DEL GITHUB

https://github.com/Magickyte/BDAutos-MongoDB